FN6802 Group Assignment 2024

Instruction: Save your answers to all questions in a PDF file and submit on NTULearn under Assignments by **<u>Tue, 10 Sep, 11:59pm</u>**.

Question 1

Monte Carlo integration is an application of the law of large numbers. Consider the integral

$$I = \int_0^1 \frac{e^{-x}}{\sqrt[3]{x}} dx.$$

1) Explain how one can evaluate I numerically using Monte Carlo integration. Write detailed steps.
2) Implement the numerical integration in Part 1) 1000 times using $n = 200$ sample points and find an appropriate way to justify the accuracy of your numerical results.
3) Instead of the uniform (0,1) distribution often used in Monte Carlo integration, one can generate a random sample of size n from a distribution with pdf
$$f_X(x) = \frac{2}{3\sqrt[3]{x}} \qquad (1)$$
on the interval (0, 1). State how to evaluate I using the sample points from this distribution.
4) Implement the numerical integration in Part 3) 1000 times with sample size $n = 200$. Evaluate the accuracy.
5) Compare two sets of 1000 estimates of I obtained by methods Parts 2) and 4), respectively, using appropriate statistical measures. Which numerical integration method do you prefer and why?

Hint: For implementation of numerical integrations in parts 2) and 4), you are required to write an R or Python code (or other computer languages that you are familiar with), which should be included in your answers together with the relevant computer outcomes.

a. To generate a size-200 random sample (i.e., 200 random numbers) from a distribution, say Uniform(0,1), in R, you can use

```
sample<-runif(200, min=0, max=1)
```

b. To generate random numbers from a distribution in (1), you need to use the probability transform as follows. Suppose that $F_X(x)$ is the cdf of the distribution of random variable X, Let $Y = F_X(X)$. Then $Y \sim Uniform(0, 1)$. To generate random numbers of X, we first generate random numbers $y_1, \cdots, y_n$ for Y from uniform(0, 1), and then random numbers for X are obtained by $x_i = F_X^{-1}(y_i)$ for $i = 1, \cdots, n$.

Question 2

To examine the occupancy records of a particular motel, a statistician chooses n dates randomly and obtains the number of rooms rented on each day as $X_1, \cdots, X_n$, which are assumed to be i.i.d. from a distribution of $N(\mu_0, \sigma^2)$ with pdf

$$f(x|\sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\{-\frac{(x-\mu_0)^2}{2\sigma^2}\},$$

where mean μ_0 is a known constant.

1) Find estimators $\tilde{\sigma}^2$ and $\hat{\sigma}^2$ for the variance σ^2 of the number of rooms rented, using the method of moments and maximum likelihood estimation, respectively.
2) Determine if estimators $\tilde{\sigma}^2$ and $\hat{\sigma}^2$ are unbiased.
3) Let $\breve{\sigma}^2 = n(\bar{X} - \mu_0)^2$, where $\bar{X} = \sum_{i=1}^{n} X_i / n$. Is $\breve{\sigma}^2$ an unbiased estimator of σ^2?
4) Among three estimators $\tilde{\sigma}^2$, $\hat{\sigma}^2$ and $\breve{\sigma}^2$, which one(s) you think is/are better for estimating of σ^2? Explain why.
5) Construct 95% confidence intervals of σ^2 using a pivot.
6) As the pivot is not unique, provide at least three different pivots for part 5). Then determine which one is the best in constructing the confidence interval of σ^2. Justify your answer both theoretically (using the knowledge learned from Lecture Notes slides 3-4) and numerically.

Hint: To numerically justify the performance of confidence intervals constructed in part 6), you are supposed to use R or Python or other computer software that you are familiar with, to do the following simulation study:

a. Simulate a random sample of size n from the $N(\mu_0, \sigma^2)$ distribution, where you can specify constants for μ_0 and σ^2, and set sample size n=100. For example, R code for simulating a random sample of size 100 from N(0, 4) is

```
s<-rnorm(100, mean = 0, sd = 2)
```

b. Then, compute three different confidence intervals that you constructed in part 6 based on the sample data you generated in s.
c. Repeat steps a)-b) 50 times. Based on these 50 random samples, compare the performance of the confidence intervals constructed by different pivots and illustrate which pivot leads to the best confidence interval using an appropriate table or graph.